

# atiyah macdonald 4

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## 1 Localization

Prop 3.10: Flatness is a local property.

Proof: Suppose  $M$  is a flat  $A$ -module, so for any exact sequence  $N' \rightarrow N \rightarrow N''$  of  $A$ -modules,  $M \otimes_A N' \rightarrow M \otimes_A N \rightarrow M \otimes_A N''$  is exact.  $M_p \otimes_{A_p} N'_p \rightarrow M_p \otimes_{A_p} N_p \rightarrow M_p \otimes_{A_p} N''_p$  is exact.

## 2 Primary Decomposition

Prop 4.1: Let  $q \triangleleft A$  be a primary ideal. Then  $r(q)$  is the smallest prime ideal containing  $q$ .

Proof: First, suppose  $xy \in r(q)$ , so  $(xy)^n = x^n y^n \in q$  for some  $n > 0$ . Since  $q$  is primary, either  $x^n \in q$  or  $y^{nk} \in q$  for some  $k > 0$ , ie  $x \in r(q)$  or  $y \in r(q)$ . Thus,  $r(q)$  is prime (we have  $r(q) \neq A$  since  $q \neq A$ ). Now suppose  $p$  is prime containing  $q$ . Then  $r(q) \leq r(p) = p$ , so  $r(q)$  is minimal.

Ex: The only primary ideals in  $\mathbb{Z}$  are 0 and  $(p^n)$  for some prime  $p$ . Proof: Let  $0 < (n) \triangleleft \mathbb{Z}$  (so  $n > 1$ ). Write  $n = \prod_i p_i^{e_i}$ , so  $r((n)) = (a)$ , where  $a = \prod_i p_i$ .  $(a)$  is not prime if  $a$  is not prime. Finally, 0 is prime and therefore primary.

Lemma 4.3: If  $q_i$  for  $1 \leq i \leq n$  are  $p$ -primary, so is  $q = \bigcap_i q_i$ .

Proof:  $r(q) = \bigcap_{i \in [n]} r(q_i) = \bigcap_{i \in [n]} p = p$ . Suppose  $xy \in q$  with  $y \notin q$ , so  $y \notin q_i$  for some  $i$ . Then  $x^n \in q_i$  for some  $n > 0$ , so  $x \in r(q_i) = p$ .

Lemma 4.4: Let  $q$  be a  $p$ -primary ideal,  $x \notin q$ . Then  $(q : x)$  is a  $p$ -primary ideal.

Proof: We have  $(q : x) \supseteq q$ , so  $r(q : x) \geq r(q) = p$ . If  $y \in (q : x)$ , then  $xy \in q$ . Since  $x \notin q$ , we have  $y \in p$ , so  $(q : x) \leq p$ . Thus,  $r(q : x) = p$ . Suppose  $ab \in (q : x)$ , so  $abx \in q$ . If  $bx \notin q$ , then  $a \in p$ .

Theorem 4.5 (First Uniqueness theorem): Suppose  $a = \bigcap_{i \in [n]} p_i$  is a minimal primary decomposition of  $a$ . Then the set of  $p_i$  equals the set of  $r(a : x)$  that are prime ideals as  $x \in A$ .

Proof: Note if  $x \in a$ , then  $(a : x) = A$ . Suppose  $x \notin a$  and denote by  $I = \{i \in [n] : x \notin p_i\} \neq \emptyset$ .

Then

$$r(a : x) = r\left(\bigcap_{i \in [n]} a_i : x\right) = \bigcap_{i \in [n]} r(a_i : x) = \bigcap_{i \in I} p_i$$

so  $r(a : x) \geq p_i$  for some  $i \in I$  (note we used  $r(q_i : x) = A$  if  $x \in q_i$ ). Conversely,  $r(a : x) \leq r(q_i : x) = p_i$ , so  $r(a : x) = p_i$ . Each  $p_i$  is achievable: pick  $x \in \bigcap_{j \neq i} q_j - q_i$ . Then  $I = \{j\}$  and  $r(a : x) = p_i$ .

### 3 Chain Conditions

Prop 6.1. Let  $(\Sigma, \leq)$  be a partial order. Then the following conditions are equivalent:

- (i). Every ascending sequence in  $\Sigma$  stabilizes.
- (ii). Every nonempty subset of  $\Sigma$  has a maximal element.

Proof: (i)  $\rightarrow$  (ii). We show the contrapositive. Let  $S \leq \Sigma$  be nonempty with no maximal element. Pick any  $x_1 \in S$ . Suppose  $x_n$  is defined. Then pick  $x_{n+1} > x_n$  in  $S$ . Suppose FTSOC the sequence  $(x_n)$  stabilizes, so there exists  $N \in \mathbb{Z}^+$  such that  $x_N = x_{N+1}$ , which contradicts the definition of  $x_n$ .

(ii)  $\rightarrow$  (i). Let  $(x_n)_{n \in \mathbb{Z}^+}$  be an ascending sequence. The set  $\{x_n\}_{n \in \mathbb{Z}^+}$  has a maximal element  $x$ , so  $x = x_N$  for some  $N \in \mathbb{Z}^+$ . For all  $n \geq N$ , we have  $x_n \leq x_N \leq x_n$ , so  $x_N = x_n$  for all  $n \geq N$ .

Ex: A finite Abelian group satisfies the acc and the dcc (follows from finite bounded monotone sequences in  $\mathbb{Z}$  must stabilize).  $\mathbb{Z}$  satisfies the acc, but not the dcc. For  $a \neq 0, \pm 1$ , we have  $(a) > (a^2) > (a^3) > \dots$ . Suppose  $A_1 \leq A_2 \leq A_3 \leq \dots$  in  $\mathbb{Z}$ . Write  $A_n = (a_n)$  for  $a_n \geq 0$ , so  $(a_n) \leq (a_{n+1}) \rightarrow a_{n+1} \mid a_n$ . If  $a_n = 0$  for all  $n$ , we're done. Else, let  $N \in \mathbb{Z}^+$  be minimal such that  $a_N \neq 0$ . Then the sequence  $(a_n)_{n \geq N}$  is a decreasing sequence in  $\mathbb{Z}_{\geq 0}$ , which stabilizes.

Prop 6.2:  $M$  is a Noetherian  $A$ -module  $\iff$  every submodule of  $M$  is finitely generated.

Proof: Let  $N \leq M$ . Let  $\Sigma$  denote the set of finitely generated submodules of  $N$  (forgot to say  $\Sigma \neq \emptyset$ ). Let  $N'$  be a maximal element of  $\Sigma$  and suppose FTSOC  $N' < N$ . Pick  $x \in N - N'$ , so  $N' < \langle N', x \rangle \leq N$ , absurd. Now suppose every  $N \leq M$  is finitely generated. Let  $(N_n)$  be an ascending sequence of modules and  $N_\infty = \bigcup_{n \in \mathbb{Z}^+} N_n$ , so  $N_\infty \leq M$ .  $N_\infty$  is finitely generated, say by  $\{x_i\}_{i \in [k]}$ . Let  $x_i \in M_{j_i}$  and  $J = \max_{i \in [k]} j_i$ . Then  $N_\infty \leq N_J \leq N_\infty$ , so  $N_\infty = N_J$ .

Prop 6.3: Suppose  $0 \rightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \rightarrow 0$  is exact. Then  $M$  is Noetherian iff  $M', M''$  are Noetherian.

Proof: We show one direction. Suppose  $M', M''$  are Noetherian. Let  $(N_n)$  be an ascending sequence of submodules of  $M$ . Let  $N'_n = \alpha^{-1}N_n$  and  $N''_n = \beta N_n$ . Suppose the sequences  $(N'_n), (N''_n)$  stabilize at  $k \in \mathbb{Z}^+$ . I claim that  $N_n$  stabilizes at  $k$ .

We wish to show that  $N_n = N_k$  for  $n \geq k$ . I claim that  $N_n = \alpha N'_n + \beta^{-1} N''_n$ . We have  $\beta^{-1} N''_n \geq N_n$  and  $\alpha N'_n \leq N_n$ . *Lysol*